

Contagion or Common Factor? Instrumental Identification and Robust Inference in High-Frequency Spillover Networks

Mauricio Herrera-Marín

Faculty of Engineering, Universidad del Desarrollo, Santiago, Chile

`mherrera@udd.cl`

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Abstract

High-frequency spillover networks are often summarized by the spectral radius of a reproduction matrix and response-based connectedness. An omitted market-wide activity factor is absorbed as a nonnegative rank-one network component, inflating both. We use two disjoint out-of-panel asset baskets as noisy proxies, one as regressor and one as instrument. The first-stage concentration parameter scales as $T\kappa^2$, so conventional linear-IV diagnostics need not protect inference on the spectral radius. We construct analytic HAC Anderson–Rubin loading sets and a certified outer confidence set for cascade risk, valid at any instrument strength under the maintained model. Monte Carlo evidence shows that plug-in 2SLS can undercover despite apparently strong first stages, while correlated proxy errors can make F large and destroy coverage. In eight one-minute cryptocurrency windows, instruments are strong in all 192 lagged designs. 2SLS reduces the spectral radius at one- and two-minute lags across all partitions, with the effect fading by five minutes and reversing at ten. Yet the joint 95% Anderson–Rubin

set intersected with the nonnegative-excitation cone is empty in every full-network design, under equal-weight and training-PC1 proxies. Robust inference therefore doubles as a specification diagnostic: the data support short-horizon common-drive correction but not the maintained pure-excitation network.

Keywords: contagion; systemic risk; spillover networks; weak instruments; Anderson–Rubin; specification diagnostics.

1 Introduction

Whether comovement across assets reflects *contagion*—shocks transmitted from one asset to another—or *interdependence* through common factors is a question with a long history in empirical finance (Forbes and Rigobon, 2002; Bekaert et al., 2005). High-frequency data have moved it from correlations of returns to *event networks*: mutually exciting point-process models in which an order-flow burst, a price jump or a liquidity withdrawal in one asset raises the intensity of similar events in others (Aït-Sahalia et al., 2015; Bacry et al., 2015). The fitted reproduction matrix B is read as a directed spillover network, its Perron root $\rho(B)$ as the systemic cascade risk or “reflexivity” of the market (Hardiman et al., 2013; Filimonov and Sornette, 2015), and the implied response matrix $(I - B)^{-1}$ as the basis for connectedness and systemic-contribution measures in the spirit of Diebold and Yilmaz (2014), Billio et al. (2012), Adrian and Brunnermeier (2016) and Acharya et al. (2017).

These readings assume that what excites events across assets is other assets’ events. In practice a market-wide activity factor—news arrival, funding-liquidity shocks, index rebalancing, the intraday clock—moves all intensities together. A self-exciting model without that factor absorbs it as if it were spillover. The companion paper (Herrera-Marín, 2026) characterizes this contamination when the factor is *observed*: the estimated reproduction matrix equals the structural one plus a rank-one term $c\gamma^\top$, where c collects the assets’ factor loadings and γ the projection of the factor on the panel’s own past; a covariate identifies the factor only through the share κ of its variation that the panel’s history does not already

explain; and cascade risk admits a finite-sample confidence set that remains valid as $\kappa \rightarrow 0$.

This paper treats the empirically relevant case: the market factor is *unobserved*. Three contributions follow.

First, we translate factor contamination into the objects that systemic-risk analysis reports. Proposition 2.1 shows that the estimated response matrix $(I - \hat{B})^{-1}$ equals the structural one plus a nonnegative rank-one term with an explicit Sherman–Morrison form, so that every pairwise response and every systemic contribution is over-stated, response connectedness can rise substantially, and rankings of systemic contribution need not be preserved. The naive network is thus not merely noisy but systematically biased toward the reading of a more interconnected, more fragile market—which is exactly the “no contagion, only interdependence” warning of Forbes and Rigobon (2002) in network form.

Second, we show that the unobserved factor can be instrumented. We treat it as a latent regressor measured with error (Wald, 1940; Fuller, 1987) and use two proxies built from *disjoint baskets of assets outside the panel*: one enters the intensity regression as a regressor, the other instruments it. Relevance is the presence of a common activity factor; exclusion is that a basket’s idiosyncratic activity does not enter the panel’s intensities except through the factor; independence of basket-specific errors is what disjointness buys. Theorem 3.2 shows that two-stage least squares (2SLS) identifies the structural network whenever $\kappa > 0$, gives the concentration parameter in closed form, and shows that it scales as $T\kappa^2$: the instrument becomes weak along sequences on which an observed factor would still be consistently estimable. With three baskets, the exclusion restriction becomes overidentified and testable (Proposition 3.3).

Third, because cascade risk is the spectral radius of a nonnegative matrix, weak-instrument distortions are amplified by the nonlinearity and by the cone projection. Conventional strength diagnostics calibrated to linear coefficients (Stock and Yogo, 2005; Montiel Olea and Pflueger, 2013) do not by themselves guarantee accurate plug-in inference for this functional: in our Monte Carlo, with a first-stage F of 28, cone-projected 2SLS retains a bias of

0.12 on a true radius of 0.70 and its delta-method interval covers 62%. We therefore construct an Anderson–Rubin confidence set ([Anderson and Rubin, 1949](#); [Kleibergen, 2002](#); [Andrews et al., 2019](#)) for cascade risk that is valid at any instrument strength (Theorem 4.1). Its structure is transparent: it is the reduced-form confidence set of the companion paper with the aliasing direction replaced by the proxy’s projection on the histories and the loading box replaced by the Anderson–Rubin set, intersected with the nonnegative cone and projected onto the spectral radius by Collatz–Wielandt duality. The Anderson–Rubin set is inverted analytically, so genuinely unbounded loading sets are reported as such rather than truncated; as the instrument weakens the confidence region expands toward the cone-restricted geometry that characterizes the non-identified observed-factor problem.

Instrumental methods for point processes exist—[Mogensen \(2023\)](#) develops instrumental processes for linear Hawkes models through integrated covariances—and the present contribution is not the use of instruments as such but their coupling with the separation-share theory that governs their strength, with the errors-in-variables structure of activity proxies, and with weak-instrument-robust inference on a spectral functional of a nonnegative matrix. On the econometric side, the closest neighbour is factor-adjusted network estimation: [Barigozzi et al. \(2024\)](#) estimate the common component of a high-dimensional panel and then the idiosyncratic network, and [Barigozzi and Brownlees \(2019\)](#) estimate sparse VAR networks in finance; those methods assume the common component can be estimated from the panel and removed, whereas here the economically relevant activity factor is latent, observed only through noisy external proxies, and potentially weakly instrumented. The setting is also the common-correlated-effects problem of [Pesaran \(2006\)](#) with lagged outcomes as regressors, where the cross-sectional-average proxy fails for a structural reason we make explicit (Remark 3.4); on the finance side it is the endogenous–exogenous decomposition of [Filimonov and Sornette \(2015\)](#) and [Wheatley et al. \(2019\)](#) at the level of the whole network rather than of a scalar branching ratio.

Recent work in this journal underscores the empirical and methodological relevance of

dynamic financial networks: [Leong \(2025\)](#) develops a direct systemic-contagion test, [Xu et al. \(2024\)](#) uses instrumental-variable quantile regression for endogenous network effects, [Barigozzi et al. \(2024\)](#) removes estimated common components before network estimation, and [Cai et al. \(2026\)](#) constructs nonparametric dynamic financial networks. Our setting differs in treating the common activity factor as latent and measured only through noisy external proxies, and in targeting identification-robust inference on the spectral radius of a constrained reproduction network.

Section 2 sets up the model in the vocabulary of spillover networks and derives the connectedness result. Section 3 gives identification and first-stage strength with two proxies. Section 4 constructs the robust confidence set. Section 5 reports the Monte Carlo evidence. Section 6 specifies the cryptocurrency application. Section 7 discusses implications for systemic-risk measurement. Proofs are in the Appendix.

2 Spillover networks, cascade risk, and factor contamination

2.1 Model and vocabulary

A panel of K assets is observed on T bins. Y_{zt} is the event count of asset z in bin t (order-flow bursts, jumps, liquidity shocks), and H_{kt} is the exponentially filtered history of asset k up to $t - 1$. The linear-projection form of a multivariate Hawkes model with a common factor f_t is

$$Y_z = \beta_z \mathbf{1} + H b_z + c_z f + \varepsilon_z, \quad \mathbb{E}[\varepsilon_z \mid H, f] = 0, \quad \text{Cov}(\varepsilon_z \mid H, f) = \sigma_z^2 I, \quad (1)$$

with ε_z independent across z . Table 1 fixes the correspondence between the statistical objects and their economic reading, which we use throughout.

Table 1: Objects of the model and their reading in spillover-network and systemic-risk analysis.

object	symbol	economic reading
reproduction matrix	$B_0 = [b_z^\top] \geq 0$	structural spillover network; $(B_0)_{zk}$ = events in z triggered per event in k
Perron root	$\rho(B_0)$	systemic cascade risk (branching ratio, reflexivity); $\rho \geq 1$ = supercritical
response matrix	$R_0 = (I - B_0)^{-1}$	total (all-generation) response; basis of connectedness measures
common factor	f	market-wide activity factor (news, funding liquidity, intraday clock)
factor loadings	$c \geq 0$	assets' exposure to the factor (activity betas)
market-memory coefficient	$\gamma = (H_c^\top H_c)^{-1} H_c^\top f_c$	how much of the factor is predictable from the panel's own past
separation share	$\kappa = \ f_\perp\ ^2 / \ f_c\ ^2$	factor-innovation share not explained by the panel's history
naive network	$A = B_0 + c\gamma^\top$	what a factor-free fit estimates (linear projection)
proxies	$m_j = \theta_j f + e_j$	activity of disjoint out-of-panel baskets, $j = 1, 2(, 3)$

2.2 Factor contamination of connectedness and systemic contributions

The companion paper's aliasing identity states that a fit of (1) without f has population coefficient $A = B_0 + c\gamma^\top$: the omitted factor is absorbed as spillover along the direction γ in which it is predictable from the histories, scaled by each asset's loading. We now carry this into the response matrix and the connectedness measures built from it. Write $R_0 = (I - B_0)^{-1}$ and $R_A = (I - A)^{-1}$, well defined when $\rho(B_0) < 1$ and $\rho(A) < 1$. For a nonnegative response matrix R , define the *response connectedness* $\mathcal{C}(R) = 100(1 - \text{tr}(D_R^{-1}R)/K)$ with D_R the diagonal of row sums—a row-normalized measure analogous to Diebold and Yilmaz (2014), constructed from the all-generation reproduction response rather than from a forecast-error variance decomposition—and the systemic contribution of asset k as its column sum $s_k(R) = \sum_z R_{zk} - 1$, the total downstream response generated by a unit event in k over all generations.

Proposition 2.1 (Rank-one contamination of the response matrix). *Suppose $B_0 \geq 0$, $c \geq 0$,*

$\gamma \geq 0$, $\rho(B_0) < 1$ and $s := \gamma^\top R_0 c < 1$. Then

$$R_A = R_0 + \frac{(R_0 c)(\gamma^\top R_0)}{1 - s}, \quad (2)$$

so that (i) $R_A \geq R_0$ entrywise, with strict inequality wherever $(R_0 c)_z(\gamma^\top R_0)_k > 0$; (ii) the systemic contribution of every asset is over-stated, $s_k(R_A) - s_k(R_0) = (\mathbf{1}^\top R_0 c)(\gamma^\top R_0)_k / (1 - s) \geq 0$, by an amount proportional to how strongly the factor's history-projection loads on k ; (iii) rankings of systemic contribution are guaranteed to be preserved when $(\gamma^\top R_0)_k$ is constant across k , and otherwise need not be preserved; (iv) $\rho(A) \geq \rho(B_0)$ and $\rho(A) \rightarrow 1$ as $s \uparrow 1$: *apparent criticality*.

The proof is a Sherman–Morrison computation (Appendix). The economic content is that a market with an unmodeled activity factor *looks* more fragile and can be differently ranked than it is: pairwise responses and systemic contributions are inflated, and the row-normalized response connectedness, while not monotone in general, can increase substantially, and that the distortion is concentrated on the assets whose past best forecasts the factor—typically the most liquid ones—which are precisely those that connectedness rankings single out as systemic. In the numerical illustration of Section 5, total connectedness rises from 38% to 75% and the two most systemic assets swap places.

3 Identification with two out-of-panel proxies

3.1 Assumptions and their economic content

Suppose f is unobserved and proxies are available.

Assumption 3.1 (Two proxies). $m_j = \theta_j f + e_j$, $j = 1, 2$, with $\theta_j > 0$, $e_j \perp (f, H, \varepsilon)$, and $e_1 \perp e_2$.

In the application m_j is the leading component of the activity of a basket of assets *outside* the panel, lagged one bin. The three parts of Assumption 3.1 are, respectively: *rel-*

evance—a market-wide activity factor loads on the basket ($\theta_j > 0$); *exclusion*—the basket’s idiosyncratic activity e_j does not enter the panel’s intensities except through f , which is the statement that there is no direct order-flow channel from the basket to the panel; and *independence of measurement errors* across baskets, which disjointness of the baskets makes plausible and which the overidentification test of Proposition 3.3 can check. Write $m_{j\perp}$ for the residual of m_j on $[\mathbf{1}, H]$ and $m^{(1)} = (H_c^\top H_c)^{-1} H_c^\top m_{1,c}$ for the proxy’s market-memory coefficient; $m^{(1)} \rightarrow_p \theta_1 \gamma$ because $e_1 \perp H$.

3.2 Attenuation, two-stage least squares, and instrument strength

Theorem 3.2 (Two proxies). *Under (1) and Assumption 3.1:*

- (a) (Attenuation.) *Least squares of Y_z on $[\mathbf{1}, H, m_1]$ is inconsistent: its H -coefficient converges to $b_z + c_z(\gamma - a m^{(1)})$ with $a = \theta_1 \|f_\perp\|^2 / (\theta_1^2 \|f_\perp\|^2 + \|e_{1\perp}\|^2) < 1/\theta_1$. A single noisy proxy removes only the fraction $a\theta_1$ of the contamination.*
- (b) (Identification.) *2SLS with m_2 instrumenting m_1 identifies $(b_z, c_z/\theta_1)$ whenever $\kappa > 0$ and is consistent along sequences with κ fixed.*
- (c) (Strength.) *The first-stage concentration parameter is*

$$\mu^2 = \frac{\pi^2 \|m_{2\perp}\|^2}{\sigma_{e_1}^2 + \theta_1^2 \text{Var}(f_\perp \mid m_{2\perp})}, \quad \pi = \frac{\theta_1 \theta_2 \|f_\perp\|^2}{\theta_2^2 \|f_\perp\|^2 + \|e_{2\perp}\|^2},$$

and, with $\|f_\perp\|^2 = \kappa \|f_c\|^2 \asymp T\kappa$, $\mu^2 \asymp T\kappa^2$ as $\kappa \rightarrow 0$.

Part (c) is the operative statement. The information about the factor available to an observer of f scales as $T\kappa$ (Herrera-Marín, 2026); the information available through an instrument scales as $T\kappa^2$. Along sequences with $T\kappa \rightarrow \infty$ but $T\kappa^2 = O(1)$ —a factor that is predictable from the panel’s past to within $O(T^{-1/2})$ —an observed factor would still be consistently separated while the instrument is weak in the sense of Staiger and Stock (1997). Instrumental identification of a spillover network is therefore *more* demanding than covariate

adjustment, not less, and the first-stage F is the diagnostic that decides which regime one is in.

Proposition 3.3 (Overidentification with three baskets). *With a third proxy $m_3 = \theta_3 f + e_3$ satisfying Assumption 3.1 jointly with m_1, m_2 , the equation for Y_z is overidentified by one; the Hansen statistic J_z (with a HAC weighting matrix; in the homoskedastic case the Sargan form TR_z^2) is asymptotically χ_1^2 under exclusion and independence of (e_2, e_3) (Sargan, 1958; Hansen, 1982). It provides an overidentification diagnostic for the joint exclusion restrictions, not a proof of validity; under weak instruments it is unreliable and the identification-robust version of Kleibergen (2005) should be used.*

Economically, J_z tests whether the second basket carries information about asset z 's intensity beyond the factor—a direct order-flow channel—which is the exclusion restriction; rejecting it for a particular asset flags a basket-to-asset linkage that the researcher must either model or avoid by choosing baskets differently.

Remark 3.4 (Why in-panel activity fails: the common-correlated-effects analogy). Pesaran (2006) proxies an unobserved common factor by cross-sectional averages of the observables. In a self-exciting network the regressors are lagged outcomes, so the cross-sectional average of lagged activity lies in $\text{span}\{\mathbf{1}, H\}$ up to the current-bin innovation: lagged it is collinear (its separation share is essentially zero), contemporaneous it is endogenous. This is why smoothed panel activity has $\kappa \approx 0.02$ in the companion paper's audits, and why the proxies must come from outside the panel.

4 Weak-instrument-robust inference on cascade risk

4.1 The Anderson–Rubin set, inverted exactly

Let $\tilde{c}_z = c_z/\theta_1$ be the loading in the scale of m_1 . For a hypothesized value c^0 , the regression of $Y_z - c^0 m_1$ on $[\mathbf{1}, H, m_2]$ has, under $H_0 : \tilde{c}_z = c^0$, error $\varepsilon_z - c^0 e_1$, uncorrelated with m_2 ;

the t -statistic of m_2 , $t_z(c^0)$, is exactly t_{T-K-2} under Gaussianity for any first-stage strength (Anderson and Rubin, 1949). After partialling $[1, H]$ out of Y_z , m_1 and m_2 , the coefficient on m_2 is affine in c^0 and the residual sum of squares is quadratic in c^0 , so $\{|t_z(c^0)| \leq t_{\text{crit}}\}$ is a quadratic inequality whose solution set is computed in closed form: a bounded interval, the real line, the complement of a bounded interval, or the empty set. We use this analytic inversion throughout; a grid inversion truncates unbounded sets at the grid edges and is therefore anti-conservative exactly when the instrument is weak. With a Newey–West variance in place of the homoskedastic one the inequality remains quadratic, because the residual is affine in c^0 ; this gives an asymptotic HAC-robust Anderson–Rubin set for the application, in the spirit of Montiel Olea and Pflueger (2013) and Andrews et al. (2019). Write $\mathcal{C}_z(\alpha')$ for the set at level α' .

4.2 From the loading to the network

Two exact facts make the passage from $\tilde{c}_z$ to the network transparent. First, for any c^0 the least-squares coefficient of $Y_z - c^0 m_1$ on H is $\hat{A}_z - c^0 m^{(1)}$, where $\hat{A}_z$ is the *naive* coefficient (short-minus-long identity): candidate networks lie on a line through the naive fit in the direction of the proxy’s market-memory coefficient. Second, under H_0 that regression is correctly specified with homoskedastic error of variance $\sigma_z^2 + (c^0)^2 \sigma_{e_1}^2$, so its H -coefficient has an exact confidence region with the estimated residual variance $s_z^2(c^0)$ and an F radius. Let $\alpha = \alpha_1 + \alpha_2$, $q_1 = K F_{K, T-K-1}(1 - \alpha_1/K)$, and let $\bar{s}_z^2$ be the largest residual variance over $c^0 \in \mathcal{C}_z(\alpha_2/K) \cap \mathbb{R}_{\geq 0}$ (the residual variance is convex in c^0 , so the maximum is at the endpoints; for an unbounded set the ellipsoid is evaluated on a bounded envelope and the cone supplies the remaining constraint).

Theorem 4.1 (Anderson–Rubin confidence set for cascade risk). *Define*

$$S_\alpha^{\text{AR}} := \left\{ A - d m^{(1)\top} \geq 0 : (\hat{A}_z - A_z)^\top (H_c^\top H_c) (\hat{A}_z - A_z) \leq \bar{s}_z^2 q_1, \quad d_z \in \mathcal{C}_z(\alpha_2/K) \cap \mathbb{R}_{\geq 0} \quad \forall z \right\}.$$

(i) (Validity.) Under (1), Assumption 3.1 and Gaussian errors, $\Pr(B_0 \in S_\alpha^{\text{AR}} \mid H, f, e) \geq 1 - \alpha$ for every $\kappa \geq 0$ and every first-stage strength, with empty sets counted as failures; the variances $\sigma_z^2, \sigma_{e_1}^2$ need not be known. (ii) (Computation.) Each $\mathcal{C}_z$ is a finite union of intervals, so S_α^{AR} is a finite union of convex sets, and $\rho(S_\alpha^{\text{AR}})$ is a union of intervals. For any $x > 0$, $\rho^+ \leq \max_{B \in S} \max_i (Bx)_i / x_i$ and $\rho^- \geq \min_{B \in S} \min_i (Bx)_i / x_i$ (Collatz–Wielandt), each a set of K convex programs; the reported interval is an outer bound of this form and inherits the coverage of S_α^{AR} . When a loading set is unbounded above and the cone does not bound d_z , the upper endpoint is reported as $+\infty$. (iii) (Strong instrument.) If κ is fixed and $T \rightarrow \infty$, $S_\alpha^{\text{AR}} \rightarrow \{B_0\}$. (iv) (Structure.) S_α^{AR} is the reduced-form set of [Herrera-Marín \(2026\)](#) with the aliasing direction γ replaced by $m^{(1)}$ and the loading box replaced by the Anderson–Rubin set: instrumental and covariate inference on cascade risk are the same object, differing only in how the loading is constrained.

The Anderson–Rubin set is the only element of the construction that depends on the instrument; everything else is the naive fit and the cone. When the instrument is weak, $\mathcal{C}_z$ is unbounded with high probability, d_z is constrained only by $d_z \geq 0$ and $B \geq 0$, and the region approaches the cone-restricted geometry that characterizes the exact-collinearity identified set of the observed-factor problem in [Herrera-Marín \(2026\)](#). We report this as what the simulations show rather than as a theorem: a formal statement would require controlling the shape of the Anderson–Rubin sets uniformly, which we do not attempt. In the terminology of [Andrews et al. \(2019\)](#), the set is robust because its validity does not depend on the concentration parameter and informative because it shrinks to the point when the parameter diverges.

Remark 4.2 (Empty sets and specification compatibility). An empty S_α^{AR} is not an unbounded weak-instrument result. It means that the rowwise loading restrictions and reduced-form confidence regions have no common element inside the maintained cone $B \geq 0$. Under the exact Gaussian conditions of Theorem 4.1, $\Pr(S_\alpha^{\text{AR}} = \emptyset) \leq \alpha$ because emptiness implies that the true B_0 is not covered. The empirical application replaces the homoskedastic loading

inversion by its HAC analogue, so we use emptiness there as an asymptotic specification-compatibility diagnostic rather than as an exact finite-sample test. In particular, an empty set is reported as $\emptyset$ and never recoded as an infinite interval.

Closed-form outer projection in the application. The confidence set itself is unchanged in the empirical computation. For a fixed Collatz–Wielandt trial vector, support of each reduced-form coefficient ellipsoid is available in closed form. We relax the nonnegativity cone only inside that support calculation, which can only enlarge the upper support and lower the lower support; the lower numerator is then truncated at zero because every feasible reproduction matrix is nonnegative. This yields a conservative certified outer interval for the same S_α^{AR} while avoiding repeated conic support solves. A single conic feasibility problem per design is retained to identify an empty intersection. If the solver fails without establishing infeasibility, the implementation returns $[0, \infty)$ rather than a false rejection. No coefficient grid or artificial cap is used.

Remark 4.3 (Benchmark). The comparison procedure in Section 5 is the *plug-in cone-projected 2SLS delta interval*: the point $\rho(\max(\hat{B}_{2\text{SLS}}, 0))$ with a delta-method variance from the unrestricted 2SLS covariance. Where the projection creates zeros the map $B \mapsto \rho(B_+)$ is not smooth, so this benchmark is deliberately the vulnerable practice that the Anderson–Rubin construction replaces.

5 Monte Carlo evidence

Design: a primitive $B_0 \geq 0$ with $\rho(B_0) = 0.70$, correlated histories, loadings $c \sim U(0.6, 1.4)$, proxies $m_j = f + e_j$; regimes indexed by the effective information $\|f_\perp\|^2 \in \{1200, 60, 8\}$ and the proxy noise $\sigma_e \in \{0.5, 0.7\}$; $\alpha = 0.05$ split equally between the ellipsoid and the Anderson–Rubin block, each Bonferroni-corrected across rows. Two audits are reported. Table 2 uses 50 replications with the full projection (each replication solves $2K(2 + n_x)$ convex programs) and reports bias of the point estimators and coverage of the projected

Table 2: Bias of cascade-risk estimators and coverage of nominal 95% intervals ($K = 5$, $T = 2000$, $\rho(B_0) = 0.70$, 50 replications). Empty sets count as failures. AR-unb.: replications in which at least one row’s Anderson–Rubin set is unbounded (analytic inversion).

regime	first-stage F	naive	OLS w/ proxy	2SLS (sd)	plug-in 2SLS delta cov.	AR cov.	AR width	AR-unb.
strong	2024	+0.29	+0.08	+0.02 (0.03)		0.90	1.00	0/50
moderate	28	+0.29	+0.25	+0.12 (0.16)		0.62	1.00	0/50
weak	1.4	+0.29	+0.28	+1.94 (3.14)		0.68	1.00	50/50

Table 3: Membership audit of the Anderson–Rubin confidence set, $\Pr(B_0 \in S_{0.05}^{\text{AR}})$, 10,000 replications per design, $T = 2000$ (Monte Carlo s.e. ≤ 0.0016). F is the median first-stage F .

design	errors	K	F	coverage
strong ($\ f_\perp\ ^2 = 1200$, $\sigma_e = 0.5$)	Gaussian	5	1990	0.9745
moderate ($\ f_\perp\ ^2 = 60$, $\sigma_e = 0.5$)	Gaussian	5	23	0.9746
weak ($\ f_\perp\ ^2 = 8$, $\sigma_e = 0.7$)	Gaussian	5	0.4	0.9763
moderate	Gaussian	10	22	0.9767
moderate	t_5 (scaled)	5	24	0.9761
moderate, $\text{corr}(e_1, e_2) = 0.3$ (violation)	Gaussian	5	323	0.0000

intervals. Table 3 audits the membership event $\{B_0 \in S_\alpha^{\text{AR}}\}$ with 10,000 replications—no convex programs are needed for membership—across additional designs, including a deliberate violation of the independence of basket errors and heavy-tailed errors. The reported projected interval, being an outer bound, inherits the membership coverage.

Five findings. (1) The unmodeled factor inflates cascade risk from 0.70 to 0.99: apparent criticality, the network-level version of [Filimonov and Sornette \(2015\)](#). (2) A single noisy proxy is nearly useless unless relevance is very strong: at $F = 28$ it removes one-eighth of the bias. (3) Strength diagnostics calibrated to linear coefficients do not by themselves guarantee accurate plug-in inference for the nonlinear cascade-risk functional: at $F = 28$, above the usual rules of thumb ([Stock and Yogo, 2005](#)), cone-projected 2SLS retains a bias of 0.12 and its delta interval covers 62% in this design; at $F = 1.4$ the point estimate is worthless. We do not claim that any $F = 28$ is insufficient; we claim that the linear-coefficient calibration does not transfer to ρ . (4) The Anderson–Rubin set covers in every projected replication and at 0.975 in the membership audit in every valid design, including $K = 10$ and heavy-tailed errors; it is conservative by construction (Bonferroni, maximal residual variance) and declares

the instrument uninformative when it is: all 50 weak-regime loading sets are unbounded. (5) The independence of basket errors is the assumption on which everything rests: with $\text{corr}(e_1, e_2) = 0.3$ the instrument *appears* strong ($F \approx 323$) and coverage is zero. A large first stage is not evidence of validity; only the overidentification diagnostic of Proposition 3.3 and the design of the baskets can speak to it. For the connectedness measures of Proposition 2.1, a representative draw with $\rho(B_0) = 0.55$ gives response connectedness 38.3% structural versus 74.8% naive, with the two most systemic assets swapping places.

6 Application: cryptocurrency spillover networks with out-of-panel baskets

6.1 Data and design

The panel contains 18 liquid Binance USDT pairs—BTC, ETH, BNB, SOL, XRP, ADA, DOGE, LTC, LINK, AVAX, TRX, DOT, ATOM, BCH, ETC, FIL, NEAR and UNI—at one-minute resolution over eight frozen windows: three stress episodes (Terra/Luna, 9–12 May 2022; FTX collapse, 8–10 November 2022; and the March 2023 banking/crypto shock, 10–13 March 2023) and five controls (8–10 August 2022, 22–24 August 2022, 10–12 October 2022, 16–18 January 2023 and 6–8 February 2023). Events are volume bursts above the within-window $q \in \{0.95, 0.97\}$ quantile; histories are exponentially filtered over 60 one-minute lags with a 10-minute half-life. Every regression also conditions on a centered cubic B-spline time basis with eight interior-knot parameters, so the basket proxies are required to explain variation beyond a flexible intraday/secular baseline. The external universe contains eight assets outside the panel (XLM, EOS, ALGO, AAVE, THETA, VET, ENJ and MANA).

The independence of basket errors is the binding assumption (Table 3), and disjointness alone does not deliver it: two baskets can share sector, liquidity or exchange-microstructure factors. The primary proxy is standardized equal-weight basket activity, which involves

no estimated loadings. Three pre-specified balanced disjoint splits of the external universe into two baskets of four are analysed in parallel. A three-way split (3/3/2) supplies the HAC Hansen J diagnostic of Proposition 3.3 by asset. The first principal component is a robustness specification: loadings are frozen from the first 30% of each window after the same within-window standardization used by the equal-weight proxy; missing aligned minute bins are mapped to the corresponding training-period mean in standardized coordinates before projection.

Timing: the main IV specifications use lags of 1, 2, 5 and 10 minutes. The falsification diagnostic is a one-minute *lead innovation*: future basket activity $m_B(t+1)$ is tested for incremental first-stage relevance for the primary exposure $m_A(t-1)$ after conditioning on H_t , contemporaneous basket activity $m_B(t)$ and the primary lag-one instrument $m_B(t-1)$. We do not use the mutual first stage of two jointly shifted future proxies as a placebo, because persistence of the common market factor can make that quantity large even without a look-ahead channel. *Inference*: first-stage strength is summarized by the conventional F and the HAC-robust effective F of Montiel Olea and Pflueger (2013); the rowwise Anderson–Rubin sets use a Newey–West bandwidth of 12 bins. No scalar F threshold authorizes plug-in inference. The equal-weight specification is primary; PC1 is a frozen robustness analysis.

6.2 Pre-registered reading

For each window, threshold, partition and lag we report κ_A, κ_B , conventional and effective F , naive/proxy/2SLS radii, the rowwise Anderson–Rubin loading-set status and the joint cone-restricted projection. Response connectedness is reported only when the fitted radius is below one, because $(I - B)^{-1}$ is not a stable all-generation response matrix otherwise. Instrument strength is classified separately from structural compatibility: a large first stage does not make an empty S_α^{AR} valid, and an empty set is never relabeled as a weak or unbounded result. The headline point-estimate statement is required to survive all three basket partitions; the four lag choices are used to determine its time profile rather than to

pool distinct horizons into a single effect.

6.3 Single-proxy baseline: a cross-fitted audit

Before instrumenting, we document what a single external proxy achieves when used as a covariate—the attenuated estimator of Theorem 3.2(a)—in the independent V6 audit included with the reproducibility package. Across 71 cross-fitted designs (six windows, subpanels of $K \in \{10, 20\}$, three thresholds and volume/volatility bursts), the external-factor loading direction is estimated on the first third of each window and frozen; all tests use the held-out two thirds. The reduction in the spectral radius from adding the proxy on top of a flexible time baseline is positive in 71/71 designs and exceeds a circular-shift null at the 5% level in 66/71; the frozen loading direction projects significantly on held-out data in 55/71. The correction is nevertheless modest: median $\Delta\rho = 0.086$ (interquartile range 0.036–0.108), while held-out Brier and log-loss improve beyond the flexible time baseline in only 41/71 designs. The time baseline alone lowers the median naive radius from 0.900 to 0.826. Thus an external proxy carries reproducible common-drive information but only partially removes it, which motivates the two-basket IV design.

6.4 Two-basket results: strength, horizon profile, and structural compatibility

The external baskets are not weak in the observed sample. Across all 192 lagged equal-weight designs, F_{eff} exceeds 10, with median 1108 and range 406–5767; in the primary S1/lag-one cells the median is 933. The three stress windows have a larger primary median (1783) than the controls (598), but both are far from the weak-first-stage regime considered in Section 5. Table 4 reports the 16 primary window/threshold cells.

The point correction is sharply horizon-dependent. At lags one and two, $\rho_{2\text{SLS}} < \rho_{\text{naive}}$ in 48/48 designs at each lag, across all three partitions; the mean reductions are 0.187 and

Table 4: Cryptocurrency spillover networks, primary equal-weight specification (split S1, lag 1). F_{eff} is the HAC-robust effective first-stage statistic; $\Delta\rho = \rho_{\text{naive}} - \rho_{\text{2SLS}}$. $S_{.05}^{\text{AR}} = \emptyset$ means that the joint Anderson–Rubin/reduced-form confidence restrictions have no intersection with the maintained nonnegative-excitation cone. “Incompat.” is the number of the 18 rowwise structural restrictions that are incompatible in the diagnostic decomposition; J rej. is the number of asset-level HAC Hansen tests rejecting at 5%.

window	q	F_{eff}	ρ_{naive}	ρ_{2SLS}	$\Delta\rho$	$S_{.05}^{\text{AR}}$	Incompat.	J rej.
Terra/Luna	0.95	4339	1.786	1.624	0.162	$\emptyset$	17/18	0/18
Terra/Luna	0.97	4405	2.005	1.915	0.090	$\emptyset$	18/18	0/18
FTX collapse	0.95	1397	1.836	1.465	0.371	$\emptyset$	15/18	1/18
FTX collapse	0.97	1444	1.867	1.584	0.283	$\emptyset$	17/18	0/18
Banking/crypto	0.95	1743	1.476	1.314	0.162	$\emptyset$	3/18	3/18
Banking/crypto	0.97	1823	1.613	1.439	0.175	$\emptyset$	8/18	2/18
Control Aug-A	0.95	667	1.602	1.391	0.211	$\emptyset$	3/18	1/18
Control Aug-A	0.97	686	1.666	1.553	0.113	$\emptyset$	2/18	1/18
Control Aug-B	0.95	1181	1.418	1.284	0.134	$\emptyset$	2/18	5/18
Control Aug-B	0.97	1241	1.355	1.266	0.088	$\emptyset$	1/18	2/18
Control Oct	0.95	567	1.380	1.121	0.259	$\emptyset$	4/18	1/18
Control Oct	0.97	623	1.461	1.287	0.174	$\emptyset$	3/18	1/18
Control Jan	0.95	406	2.137	1.883	0.254	$\emptyset$	16/18	0/18
Control Jan	0.97	432	2.050	1.790	0.261	$\emptyset$	15/18	1/18
Control Feb	0.95	554	1.620	1.442	0.178	$\emptyset$	7/18	2/18
Control Feb	0.97	574	1.586	1.425	0.161	$\emptyset$	5/18	1/18

0.107. At lag five the reduction remains positive in 42/48 designs but averages only 0.014. At lag ten it is positive in only 4/48 designs and the mean difference reverses to -0.033 . In the primary S1/lag-one cells, 2SLS reduces the radius in 16/16 cases by 0.192 on average (11.4% relative), but every corrected radius remains above one. The data therefore support a short-horizon common-drive correction, not a horizon-invariant reduction in propagation.

The robust set changes the interpretation of these point estimates. The rowwise HAC Anderson–Rubin loading sets are bounded in every main design, yet the joint 95% set $S_{.05}^{\text{AR}}$ is empty in all 192 full-network designs. There are no nonempty intervals with an infinite upper endpoint. A rowwise decomposition of the 16 primary cells (18 assets each) identifies 136/288 incompatible row-designs: 78/108 (72.2%) in stress windows and 58/180 (32.2%) in controls. The median factor by which the reduced-form ellipsoid radius would need to expand before intersecting the nonnegative cone is 1.47 in stress windows and 0.71 in controls. The

tension is distributed rather than driven by one removable node: DOGE, BTC and ETH are incompatible in 11/16 primary cells, BNB in 10/16, and every asset is incompatible in at least four. Thus the empty sets are not a weak-IV artifact; they diagnose incompatibility between the instrument-based loading restrictions and a full-network model constrained to purely nonnegative excitation. Once that joint set is empty, the 2SLS radii are informative point diagnostics of the direction and timing of the common-drive correction, but they are not reported as a validated structural confidence statement for $\rho(B_0)$.

6.5 Robustness and exclusion diagnostics

The PC1 robustness run reproduces the point results almost exactly. Across the 192 lagged cells, the absolute difference between PC1 and equal-weight ρ_{2SLS} has median 0.0010, mean 0.0017 and maximum 0.0088; all 192 PC1 joint AR sets are again empty. Hence neither the short-horizon point correction nor the structural incompatibility is an artifact of equal weighting.

Exclusion diagnostics are less uniformly reassuring than first-stage strength. For the corrected lead-innovation placebo, the median F_{eff} is 5.82 and 12/48 equal-weight cells exceed 10 (13/48 with PC1), although the placebo/main-lag-one F_{eff} ratio has median only 0.63% and maximum 2.77%. The equal-weight HAC Hansen diagnostic rejects at 5% in 21/288 asset-designs (7.3%) and at 10% in 48/288 (16.7%); PC1 gives 28/288 (9.7%) and 47/288 (16.3%), respectively. These diagnostics do not establish a direct basket-to-panel channel, but neither do they justify treating exclusion as empirically settled. Together with the empty joint sets, they favor a conservative reading: the external baskets reveal a strong short-horizon common component, while the pure-excitation structural network is too restrictive for the full 18-asset system.

7 Discussion

The paper reframes a question of network science as one of applied econometrics. A spillover network estimated from high-frequency events without the market factor is not merely noisy: the omitted factor creates a structured low-rank distortion. The instrumental design makes the source of identifying information explicit, and the $T\kappa^2$ strength law shows why noisy proxies are more demanding than direct covariate adjustment. The Monte Carlo adds a second lesson: first-stage strength and validity are different questions, and inference on a nonlinear spectral functional can fail even when a linear-IV diagnostic looks reassuring.

The empirical application adds a third lesson that was not imposed by the theory. The external baskets are extraordinarily relevant and the 2SLS point correction is stable across partitions at one- and two-minute horizons, but it fades with lag. At the same time, the confidence construction does not return an imprecise structural interval: it returns an empty intersection with the nonnegative-excitation cone in every full-network design. Because the rowwise loading sets are bounded, this is not the geometry of a weak instrument. It is a specification-compatibility result. The fact that the same outcome survives a PC1 proxy and is distributed across assets makes a one-node or one-weighting explanation implausible. In that sense the robust procedure does more than widen standard errors: it can expose when a convenient structural network class is inconsistent with the instrument-based restrictions.

The result also limits the substantive financial claim. We find strong evidence for a short-horizon common-drive correction to naive propagation estimates, but we do not infer a uniquely identified nonnegative structural network or a subcritical corrected market. All primary 2SLS radii remain above one, so stable all-generation response connectedness is not defined in those cells. The lead-innovation and Hansen diagnostics show some residual exclusion tension as well. Accordingly, the application should be read as evidence that common market activity contaminates short-horizon spillover estimation and that a pure-excitation 18-asset model is too restrictive, not as a causal decomposition of every observed edge.

Several extensions follow directly. A multi-factor common component would replace $c\gamma^\top$ by a low-rank term and require additional external proxies. Signed effective interactions could relax $B \geq 0$, but doing so after observing the empty cone-restricted set would change both the Hawkes interpretation and the pre-specified model; we therefore do not retrofit that extension here. Time-varying proxy loadings, richer market-microstructure controls and identification-robust tests tailored to multiple noisy proxies are natural next steps. More generally, the empty-set result suggests that specification diagnostics deserve to be reported alongside point network estimates in empirical systemic-risk work.

Supplementary materials

Reproducibility package: The archive contains the analytic/HAC Anderson–Rubin, 2SLS, Hansen and certified-projection code; the Monte Carlo generators and membership audit; the final application notebook; the script that regenerates Table 4; the single-proxy V6 audit; frozen equal-weight and PC1 outputs; rowwise structural-compatibility diagnostics; and SHA-256 checksums. Raw Binance trades are not redistributed; the archive records the public-data pipeline and the exact frozen derived outputs used in the manuscript.

Data and code availability

A versioned public archive containing the code, frozen outputs and checksums will be deposited before submission; its DOI will be inserted in the final submission files. The raw market data are publicly obtainable from Binance and are not redistributed in the archive.

Disclosure statement

The author reports no conflict of interest.

A Proofs

Proposition 2.1. $I - A = (I - B_0) - c\gamma^\top$; by the Sherman–Morrison formula, $(M - uv^\top)^{-1} = M^{-1} + M^{-1}uv^\top M^{-1}/(1 - v^\top M^{-1}u)$ for invertible M with $v^\top M^{-1}u \neq 1$; take $M = I - B_0$, $u = c$, $v = \gamma$. Nonnegativity of R_0 (Neumann series, $\rho(B_0) < 1$), of c and of γ gives (i) and (ii); (iii) is immediate from (ii); (iv) follows from $A \geq B_0$ entrywise and Perron–Frobenius monotonicity, and from the secular equation $\gamma^\top(\lambda I - B_0)^{-1}c = 1$ whose root tends to 1 as $s \uparrow 1$.

Theorem 3.2. (a) Substituting $f = (m_1 - e_1)/\theta_1$ the error becomes $\varepsilon_z - (c_z/\theta_1)e_1$, correlated with m_1 through e_1 ; by Frisch–Waugh–Lovell the coefficient on $m_{1\perp}$ has probability limit $c_z a$ with a as stated, and the short-minus-long identity transfers the shortfall $c_z(1 - a\theta_1)$ to the H -coefficient along $m^{(1)}$. (b) $m_2 \perp (e_1, \varepsilon_z)$ gives exclusion; relevance holds iff $\text{Cov}(m_{1\perp}, m_{2\perp}) = \theta_1\theta_2\|f_\perp\|^2 \neq 0$, i.e. $\kappa > 0$; consistency of 2SLS under fixed κ is standard. (c) The first-stage coefficient of $m_{2\perp}$ is π as stated; the first-stage residual variance is $\sigma_{e_1}^2 + \theta_1^2 \text{Var}(f_\perp \mid m_{2\perp})$; the concentration parameter is $\pi^2\|m_{2\perp}\|^2$ divided by it. As $\kappa \rightarrow 0$ with $\|e_{2\perp}\|^2 \asymp T\sigma_{e_2}^2$, $\pi \asymp \theta_1\theta_2\|f_\perp\|^2/(T\sigma_{e_2}^2)$ and $\|m_{2\perp}\|^2 \asymp T\sigma_{e_2}^2$, so $\mu^2 \asymp \theta_1^2\theta_2^2\|f_\perp\|^4/(T\sigma_{e_1}^2\sigma_{e_2}^2) \asymp T\kappa^2$ using $\|f_\perp\|^2 \asymp T\kappa$.

Proposition 3.3. Standard; see [Sargan \(1958\)](#) for the homoskedastic TR^2 form, [Hansen \(1982\)](#) for the HAC-weighted J used in the application, and [Kleibergen \(2005\)](#) for the identification-robust version. A J that does not reject is a diagnostic for the joint exclusion restrictions, not a proof of validity.

Theorem 4.1. (i) At the true $\tilde{c}_z$, the regression of $Y_z - \tilde{c}_z m_1$ on $[\mathbf{1}, H]$ has error $\varepsilon_z - \tilde{c}_z e_1$, independent of H (Assumption 3.1) and Gaussian, so its H -coefficient $\hat{b}_z(\tilde{c}_z) = \hat{A}_z - \tilde{c}_z m^{(1)}$ satisfies $(\hat{b}_z(\tilde{c}_z) - b_z)^\top (H_c^\top H_c)(\hat{b}_z(\tilde{c}_z) - b_z)/(\sigma_z^2 + \tilde{c}_z^2 \sigma_{e_1}^2) \sim \chi_K^2$ exactly; with the residual variance estimated, the standardized quadratic form is exactly $KF_{K, T-K-1}$ under Gaussianity; replacing $s_z^2(\tilde{c}_z)$ by $\bar{s}_z^2 \geq s_z^2(\tilde{c}_z)$ on the event $\tilde{c}_z \in \mathcal{C}_z$ can only enlarge the region. The AR event $\{\tilde{c}_z \in \mathcal{C}_z(\alpha_2/K) \forall z\}$ has probability $\geq 1 - \alpha_2$ by exact pivotality and Bonferroni across rows; the ellipsoid event has probability $\geq 1 - \alpha_1$; on their intersection $B_0 = A - \tilde{c} m^{(1)\top}$

with $A_z = \hat{A}_z - (\hat{b}_z(\tilde{c}_z) - b_z)$ lies in S_α^{AR} because $\tilde{c} \geq 0$ and $B_0 \geq 0$. Bonferroni across the two blocks gives $1 - \alpha$. (ii) Each AR set is a finite union of intervals, so the product is a finite union of boxes; on each box the set is $\{A - dm^{(1)\top} \geq 0 : A \in E_A, d \in \text{box}\}$, linear in (A, d) with convex constraints. The outer bounds follow from the Collatz–Wielandt characterization $\rho(B) = \inf_{x>0} \max_i (Bx)_i/x_i = \sup_{x \geq 0} \min_{i:x_i>0} (Bx)_i/x_i$ for $B \geq 0$ (Horn and Johnson, 2013), valid for every x and hence for any finite set of trial vectors. (iii) With κ fixed, $\mathcal{C}_z$ shrinks to $\{\tilde{c}_z\}$ at rate $T^{-1/2}$ and E_A to $\{A\}$. (iv) By inspection of the definitions.

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